



School of Physics, Engineering and Technology Assessment 2024/25

Module Code

ELE00063I

Module Name

Audio Technology and Psychoacoustics

Assessment Name

Synthesis & Psychoacoustics Exercise

% contribution

100

Accepted submission formats

One file in one of the following accepted file formats:

Multiple files in one or more of the following accepted file formats:

Single Portable Document Format (PDF) file

ZIP file containing your Pd code and any associated sound files and/or visual media

Submission is via VLE.

Please try and submit early as late submissions will be penalised.

Academic Integrity

It is your responsibility to ensure that you understand and comply with the University's policy on academic integrity. If this is your first year of study at the University then you also need to complete the mandatory Academic Integrity Tutorial. Further information is available at <http://www.york.ac.uk/integrity/>.

In particular please note:

- Unless the coursework specifies a group submission, you should assume that all submissions are individual and should therefore be your own work.
 - All assessment submissions are subject to the Department's policy on plagiarism and, wherever possible, will be checked by the Department using Turnitin software.
-

Audio Technology and Psychoacoustics – Assignment 2024/25

BRIEF INTRODUCTION AND CONTENTS

Welcome to your assignment for *Audio Technology and Psychoacoustics*.

Your assignment is to make a **single Pd patch** which has **two distinct purposes**:

- a **creative synthesiser** with an **audio-visual composition**, and
- an **interactive demonstration of a psychoacoustic phenomenon**.

You will describe your work in a single report, submitted as a PDF file. You will receive a single feedback sheet with marks and comments for all the sections.

This document contains the following subsections, which set the scene from a fictitious company, then specifies your deliverables in terms of:

- a synthesiser,
- a psychoacoustic demonstration, and
- a report.

CONTENTS

1. **Cover letter from *SynthAcademy***
2. **Synthesiser Pd Patch**
3. **Psychoacoustic Demonstration**
4. **Report**
5. **Marking Scheme**

1. COVER LETTER FROM *SYNTHACADEMY*



Subject: Exciting New Project for Aspiring Audio Software Developers - Electronics in Schools Initiative

From: Max Volume, Head of Audio Development, SynthAcademy Inc.

To: Junior Audio Software Developer

Dear new starters,

Congratulations on joining *SynthAcademy* as a Junior Audio Software Developer!

We are thrilled to have you on board and believe that your academic background in audio engineering will be a great asset to our team.

As you may know, the UK Government has recently launched a nationwide educational strategy called *Electronics in Schools*, which will be implemented from 2025. This initiative aims to raise the profile of engineering, programming, and design for all secondary and sixth-form students. One of the key subsections of the curriculum is "Audio Synthesis," and we, at *SynthAcademy*, are proud to be at the forefront of this educational movement, having won one of the contracts for development.

In response to this initiative, we have an exciting new project for you that aligns with our company's mission of making music technology accessible to a wider range of students. Your task is to create a single piece of software using the *Purr Data* programming environment, designed specifically for the Audio Synthesis subsection of the *Electronics in Schools* curriculum.

This software is aimed at sixth-form Music Technology A-level students and will have three main components:

1) A Synthesiser with an Intuitive User Interface:

Your first task is to design a synthesiser in *Purr Data* with a clear and user-friendly interface that allows sixth-form students to experiment with synthesis easily. The goal is to create an engaging learning experience that not only sparks their interest in the world of audio engineering but also aligns with the learning objectives of the *Electronics in Schools* initiative. You are to choose a synthesiser and build it to a high quality with an intuitive interface. To showcase the capabilities of this synthesiser, create an audio-visual composition using the *GEM* graphics framework. This composition should demonstrate how the synthesiser can be used for creative expression, as well as its technical capabilities.

2) Demonstration of a Psychoacoustic Phenomenon:

The same *Purr Data* software must include a demonstration of a psychoacoustic phenomenon of your choice. This demonstration should include interactive elements to allow the user to explore the chosen phenomenon. This feature will help students understand the underlying principles of sound and perception, and how these principles can be applied in audio engineering. Ensure that the demonstration is both informative and engaging, as it will play a crucial role in capturing the students' interest and meeting the goals of the *Electronics in Schools* curriculum.

3) A technical and analytical Report:

To complete the project, you are required to submit a **single report** detailing the development process, the features of the synthesiser, the chosen psychoacoustic phenomenon, and how the software aligns with the Audio Synthesis objectives of the *Electronics in Schools* initiative. This report will be used for internal evaluation, as well as a showcase for potential partnerships with educational institutions and government agencies.

We are confident that your skills and passion for audio engineering will lead to a successful and innovative piece of software that contributes significantly to the *Electronics in Schools* initiative. We look forward to reviewing your work and incorporating it into our educational product line.

Please feel free to reach out if you have any questions or require further clarification.

Best regards,

Max Volume

Head of Audio Development
SynthAcademy Inc.

Further Assessment Guidance

2. SYNTHESISER PD PATCH

First, a reminder of the synthesis task briefing from *SynthAcademy*:

A Synthesiser with an Intuitive User Interface

Your first task is to design a synthesiser with a clear and user-friendly interface that allows sixth-form students to experiment with synthesis easily. The goal is to create an engaging learning experience that not only sparks their interest in the world of audio engineering but also aligns with the learning objectives of the Electronics in Schools initiative. You are to choose a synthesiser and build it to a high quality with an intuitive interface. To showcase the capabilities of this synthesiser, create an audio-visual composition using the GEM graphics framework. This composition should demonstrate how the synthesiser can be used for creative expression, as well as its technical capabilities.

Breaking this down into its components, we have:

- Design a synthesiser in Purr Data,
- Allow sixth-form students to experiment with synthesis easily,
- Create an engaging learning experience,
- Choose a synthesiser,
- Build it to a high quality,
- With an intuitive interface,
- Create an audio-visual composition using the GEM graphics framework to demonstrate the synthesiser's use for creative expression.

Let's expand each of these points, to explain what the brief is looking for.

2.1 Design a synthesiser in Purr Data

Use the latest stable version of the *Purr Data* audio-visual programming language to create a coherent synthesiser. As you are planning your design, think carefully about how your single Pd patch can act as both an audio synthesiser and an interactive psychoacoustic demonstrator (see Section 3). The skill is to design a patch that allows people to play a synthesiser, and to experiment with a psychoacoustic phenomenon.

2.2 Allow sixth-form students to experiment with synthesis easily

Even though your brief is from a company (*SynthAcademy*) please note that your target users are sixth-form students who are studying Music Technology at A-level. You may have experience from taking such a qualification but do some research into what students at this level need to know about synthesis. Use this to inform your design decisions. The language in your interface should be aimed at 6th formers, not audio industry professionals.

2.3 Create an engaging learning experience

Design your synthesiser so that it can be used in lessons or for self-study. You might want to produce an example lesson plan (to include as an Appendix). You can expect that students will want to a) experiment with the synth, b) explore the patch (and any sub-patches) to discover how it works, c) use it for creative purposes (e.g., in a multitrack recording), d) learn how your chosen synthesis method works. Think about these things as you are planning the layout of your Pd patch.

2.4 Choose a synthesiser

Although many synthesis methods are available, you are only expected to select one synthesis method. Select a method (e.g., additive, subtractive, FM, wavetable, Karplus Strong etc.) that will lend itself to the above requirements. Do some research on its background and origin, and how it is currently used in hardware and software synthesisers.

2.5 Build it to a high quality

Build one synthesiser and do it well. Investigate what are the most useful features about your chosen synthesis method. Make sure it sounds good! Make sure that there are enough controls over the synth that people can explore a range of timbres. Add on any appropriate effects (such as reverb, chorus, or delay) to ensure that the sound could be used in a recording without needing external effects. Implement some *presets* so that the user can quickly hear what is possible with the synthesiser.

2.6 With an intuitive interface

Think about all the things that make for an *attractive* and *intuitive* interface. Those are different things. Attractive appeals to a user's first glance and involves colour, layout, labelling, neatness, professionalism etc. Intuitive means that a user doesn't need an external instruction manual to operate. It involves sensible layout, following user-interface guidelines, several rounds of testing and feedback, labelling and help instructions, quick-start options for the user etc.

2.7 Create an audio-visual composition using the GEM graphics framework

This is a chance to do something original and unusual. The request is that you demonstrate your synth with a composition - **ideally around 1 minute in length** (minimum 30s, maximum 2 minutes).

You should use Pd's GEM framework to provide a visual counterpart to your audio composition. You can think of this either as a 'music visualiser' where aspects of the sound, notes, volume and timbre control visual objects on the screen.

Alternatively, you can plan a freeform visual piece that runs alongside the synth demo, but you should consider carefully how it complements the audio.

Take some time to study the history and background of audio-visual composition, and refer to this in your documentation.

3. PSYCHOACOUSTIC DEMONSTRATION

Demonstration of a Psychoacoustic Phenomenon

The same *Purr Data* software must include a demonstration of a psychoacoustic phenomenon of your choice. This demonstration should include interactive elements to allow the user to explore the chosen phenomenon. This feature will help students understand the underlying principles of sound and perception, and how these principles can be applied in audio engineering. Ensure that the demonstration is both informative and engaging, as it will play a crucial role in capturing the students' interest and meeting the goals of the Electronics in Schools curriculum.

Breaking this down into its components, we have:

- implement a demonstration of a psychoacoustic phenomenon in *Purr Data*,
- design an interactive and engaging interface,
- demonstrate the underlying psychoacoustic principles.

Let's expand each of these points, to explain what the brief is looking for.

3.1 Implement an interactive demonstration of a psychoacoustic phenomenon in Purr Data

In the lecture and lab sessions for this module we will look at several psychoacoustic phenomena. A list of suggestions will be provided, but you are free to choose whatever phenomenon you would like to investigate. Your finished patch should successfully and clearly demonstrate your chosen phenomenon. The user should be able to interact with the patch to produce variation in the demonstration. You should think about how you can make use of automation, scripting, and randomisation to provide a wide variety of examples of the phenomenon under test.

3.2 Design an engaging interface

As covered in Section 2.6, you should design the interface of your patch to be intuitive and attractive.

3.3 Communicate the underlying psychoacoustic principles

As well as demonstrating your chosen phenomenon, your patch should also allow the user to make sense of the underlying psychoacoustic principles and therefore understand what is going on. To achieve this, your patch should be organised into appropriate sub-patches and be clearly laid out. You should ensure every part of your designed patch is self-commenting. You should also make sure that adding in this detail does not interfere with the effectiveness of the demonstration.

4. REPORT

Write a technical report to explain the design and construction of your synthesiser, and to show evidence of user testing and evaluation.

It should also explain the design of your interactive psychoacoustic demonstrator with reference to relevant academic literature.

You should document your use of time- and frequency- domain analysis tools to relate the properties of the produced audio signal to the percepts of the phenomenon.

4.1 Report Formatting

We would like you to use the IEEE conference template to produce your report, as this is an international conference standard. This also has the by-product of ensuring that your report is neat and well-structured.

This is an opportunity to explore the LaTeX document preparation system, for which a modified template is provided.

However, you can also download the Word templates from here:

<https://www.ieee.org/conferences/publishing/templates.html>

4.2 Writing Style

Write your report in 3rd person present tense
e.g. "The Pd patch is split into three main sections".

Do not use 1st person diary-style, such as:
"I made my patch first, then I tested it until I was happy with it",
because technical reports are about the product, not the author.

This will help you to write all future reports in a professional way, increase your readability, and improve the academic quality and acceptability of your work.

Make sure to use spelling and grammar checkers, and to carefully proofread your report before submission.

4.3 Report Structure

We recommend that you use the structure given on the next page.

We use italics to explain the expected content of each section.

Please also note the page limit to the main sections.

Structure (with maximum page lengths indicated for each section)

- **Title**
- **Contents**
- **1. Introduction (1 page)**
(Overview of report contents; Response to the briefing; How your approach combines synthesis and psychoacoustics)
- **2. Synthesis Report (4 pages)**
 - 2.1 Introduction to your synthesiser
(Synth method; theoretical background; context; how you've made your patch 'self-documenting')
 - 2.2 Pd design rationale
(Structure of your Pd patch and any sub-patches; how you've achieved the main synthesiser build - use annotated screenshots)
 - 2.3 Audio-visual demonstration
(Aesthetic choice of visual objects in relation to the synth audio; how you've used GEM to build the composition; linking with the synth)
 - 2.4 Evidence of testing and evaluation
(Your testing plan to show how well each section works; qualitative feedback from user tests)
- **3. Psychoacoustics Report (4 pages)**
 - 3.1 Explanation of the chosen psychoacoustic phenomenon
(Context of relevant fundamentals of psychoacoustics; description of the phenomenon; references to academic literature)
 - 3.2 Design and implementation of the Pd patch
(Overview of patch signal flow; use of automation; user interface design)
 - 3.3 Evaluation of the effectiveness of the finished patch
(Quality of the demonstration produced; ease-of-use; interactivity)
 - 3.4 Time- or frequency-domain analysis of the patch output audio
(Suitable use of waveform plots, spectral plots, spectrograms; appropriate axes; labels/titles etc; accompanying explanation)
- **4. Conclusion (1 page)**
(Refer to briefing and summarise your response; overall evaluation of exercise including future work ideas)
- **5. References**
- **Appendices (Optional)**

5. MARKING SCHEME

- **10%** Report presentation, structure, and readability
- **5%** Quality of References
- **5%** Integration of Synthesis and Psychoacoustics
- Synthesis
 - **10%** Sound quality of chosen synthesis method
 - **10%** Quality and creativity of AV demonstration
 - **10%** Pd patch presentation, and documentation
 - **10%** User interface design, documentation, and clarity.
- Psychoacoustics
 - **10%** Success of the psychoacoustic demonstration
 - **10%** Interactivity of the psychoacoustic demonstration and Pd patch quality
 - **10%** Written explanation of the psychoacoustic demonstration (with reference to relevant literature)
 - **10%** Time- and frequency-domain analysis of psychoacoustic demonstration audio.